

Sec 2.3 – The Derivative Function

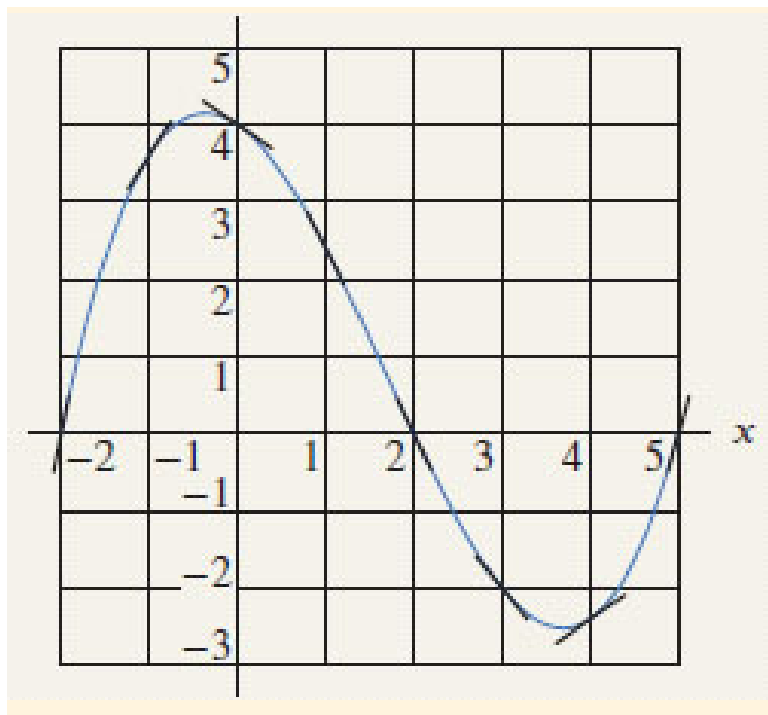
(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

In Sec 2.2 we considered the derivative of a function at a single point. Now we consider the derivative of a function at a variety of points. It generally takes on different values at different points and is itself a function.

Geometrically we can think of the derivative as the slope of the curve (or of the tangent line) at a given point.

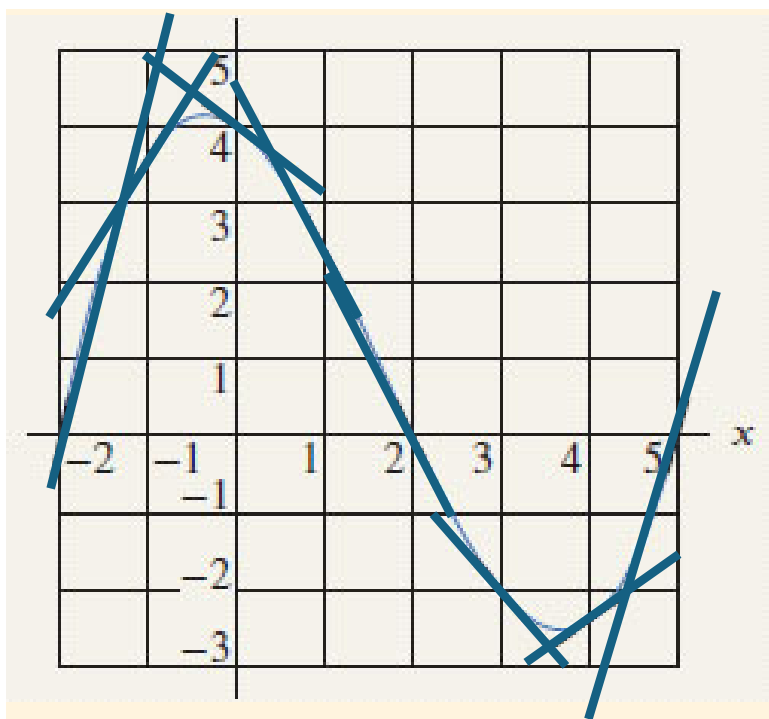
Partner up

Estimate the derivative of the following function at $x = -2, -1, 0, 1, 2, 3, 4, 5$.



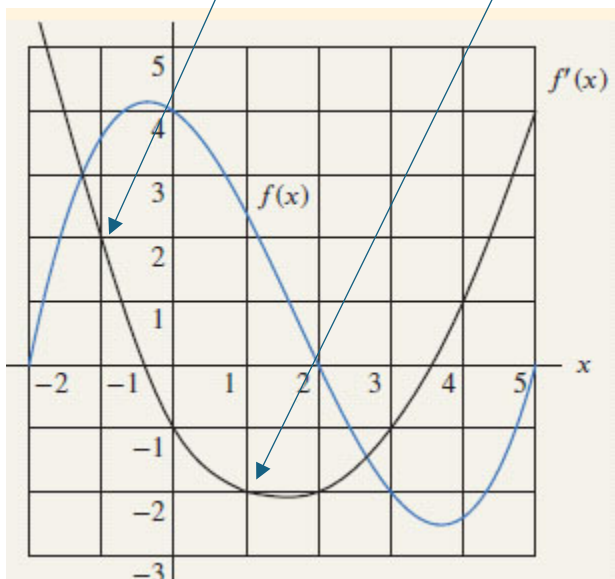
Note: use a straight-edge and draw the tangent at each point, then use the grid lines to estimate the slope.

x	-2	-1	0	1	2	3	4	5
$f'(x)$								



x	-2	-1	0	1	2	3	4	5
$f'(x)$	5 (textbook says 6, remember, these are estimates)	2	-1 (note the negative sign)	-2 (note the negative sign)	-2 (note the negative sign)	-1.3 (textbook says 6, remember, these are estimates)	1	4

Now, sketch the graph of the derivative function, using the values you calculated in the table above.



Does this graph of f' make sense? When the values of f' are positive, f is increasing, and when f' is negative, f is decreasing. The larger values of f' correspond to where f has steeper slopes.

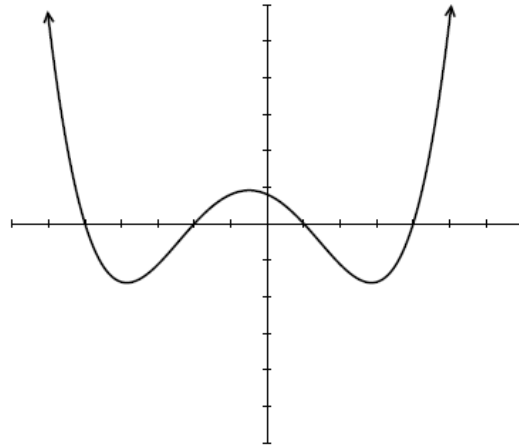
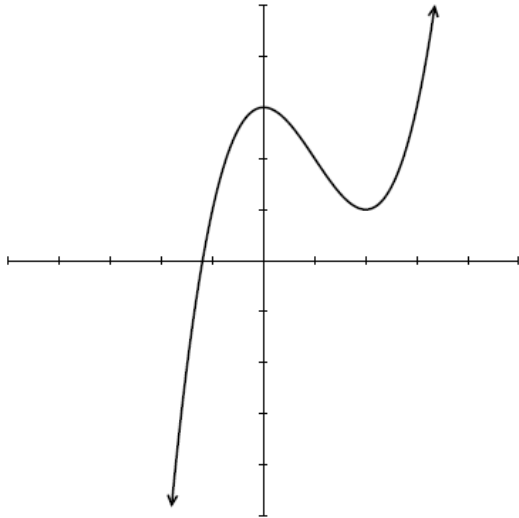
So, we formalize this with:

If $f' > 0$ on an interval, then f is *increasing* over that interval.

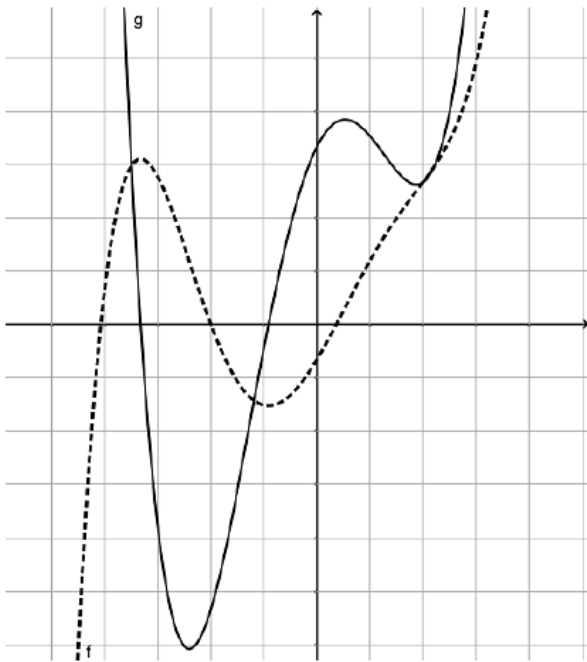
If $f' < 0$ on an interval, then f is *decreasing* over that interval.

Partner up

Example 1. Given the graph of the function f below, sketch the graph of the derivative of f in each case.



Example 2. One of the functions graphed below is the derivative of the other. Determine which is the original function and which is the derivative.



Finally, we will estimate the derivative numerically.

Values of $f(x)$ are in the table. Where in the interval $-12 \leq x \leq 9$ does $f'(x)$ appear to be the greatest? Least?

x	-12	-9	-6	-3	0	3	6	9
$f(x)$	1.02	1.05	1.12	1.14	1.15	1.14	1.12	1.06

x	-12	-9	-6	-3	0	3	6	9
$f(x)$	1.02	1.05	1.12	1.14	1.15	1.14	1.12	1.06

0.03 0.07 0.02 0.01 -0.01 -0.02 -0.06

largest
(most positive)

smallest
(most negative)